

Group 7 Report: Joint Denoising and Low-Light Image Enhancement with DiffLL

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1 Overview

1.1 DiffLL

Low-light image enhancement remains a challenging problem in computer vision due to the severe degradation caused by insufficient illumination, including loss of details, low contrast, and amplified sensor noise. Early approaches primarily relied on histogram equalization techniques and Retinex theory, which models an image as the decomposition of illumination and reflectance components. By separately estimating these two factors, Retinex-based methods aim to recover a visually enhanced image under normal lighting conditions.

With the rapid development of deep learning, convolutional neural network (CNN)-based methods [6], followed by generative models such as Generative Adversarial Networks (GANs) [3] and Variational Autoencoders (VAEs) [4], have achieved substantial improvements in low-light image enhancement. More recently, diffusion models [1] have demonstrated state-of-the-art performance across a wide range of image restoration and generation tasks. The core idea of diffusion models is to progressively corrupt data with Gaussian noise and learn a multi-step denoising process that reconstructs the clean signal.

1.2 Test-time augmentation

Test-Time Augmentation (TTA) [5] is a widely adopted technique in image classification. Instead of relying on a single input during inference, TTA applies multiple data augmentations to the test image and aggregates the predictions from all augmented samples. This strategy generally improves prediction accuracy and robustness by reducing sensitivity to input variations. Furthermore, the consistency among predictions from different augmentations can provide a useful estimate of model uncertainty.

Common TTA operations include cropping, shifting, scaling, rotation, and horizontal or vertical flipping. More advanced variants additionally incorporate photometric transformations, such as brightness and contrast adjustments, to further improve model generalization under diverse imaging conditions. By exploiting the invariance of model predictions to these transformations, TTA can effectively enhance inference performance without requiring additional training or modifications to the network architecture.

Our code is available at <https://github.com/JJW0220-gif/Diffusion-Low-Light>.

2 Methodology

2.1 Problem Statement

Given a paired low-light image enhancement dataset, the objective is to learn a restoration function that simultaneously enhances image brightness and suppresses noise. The dataset consists of 30,000 paired low-light and normal-light image patches for training, 1,000

paired patches for validation, and 1,000 low-light image patches for testing. Each image patch has a spatial resolution of 256×256 with three RGB channels.

Let $x \in \mathbb{R}^{256 \times 256 \times 3}$ denote a low-light image patch and $y \in \mathbb{R}^{256 \times 256 \times 3}$ denote its corresponding ground-truth normal-light image patch. The enhancement model G is trained in a supervised manner using the paired training set to learn the mapping

$$\hat{y} = G(x),$$

where $\hat{y}$ represents the restored image produced by the model.

During training, model parameters are optimized using the training set, while hyperparameters are selected based on performance on the validation set. The learned model is subsequently evaluated on the unseen test set.

Following common practices in low-light image enhancement benchmarks, restoration quality is assessed using three widely adopted metrics: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS). Higher PSNR and SSIM values indicate better reconstruction fidelity and structural consistency, whereas lower LPIPS values correspond to superior perceptual quality.

In addition to restoration performance, computational efficiency is also considered. Specifically, model complexity is evaluated in terms of the number of trainable parameters (M Params) and floating-point operations (FLOPs). Therefore, the goal is to develop an enhancement model that achieves high PSNR and SSIM, low LPIPS, and low computational cost simultaneously, thereby providing an effective balance between restoration quality and efficiency.

2.2 Baseline

DiffLL [2] is a diffusion-based low-light enhancement framework that leverages the Haar wavelet transform to perform lossless frequency decomposition. Specifically, an input image is decomposed into one low-frequency approximation coefficient and three high-frequency detail coefficients, corresponding to the horizontal, vertical, and diagonal frequency components. To further reduce computational complexity, the low-frequency approximation coefficient is recursively decomposed through K levels of wavelet transformation, while all high-frequency coefficients are preserved for subsequent reconstruction. Diffusion modeling is then applied only to the low-dimensional low-frequency representation, significantly reducing inference cost. Furthermore, the low-frequency coefficients extracted from the low-light image are utilized as guidance during the denoising process.

To restore high-frequency details, DiffLL introduces a High-Frequency Restoration Module (HFRM). The high-frequency coefficients are first separated into RGB channels. Cross-attention is then employed between the horizontal and vertical high-frequency

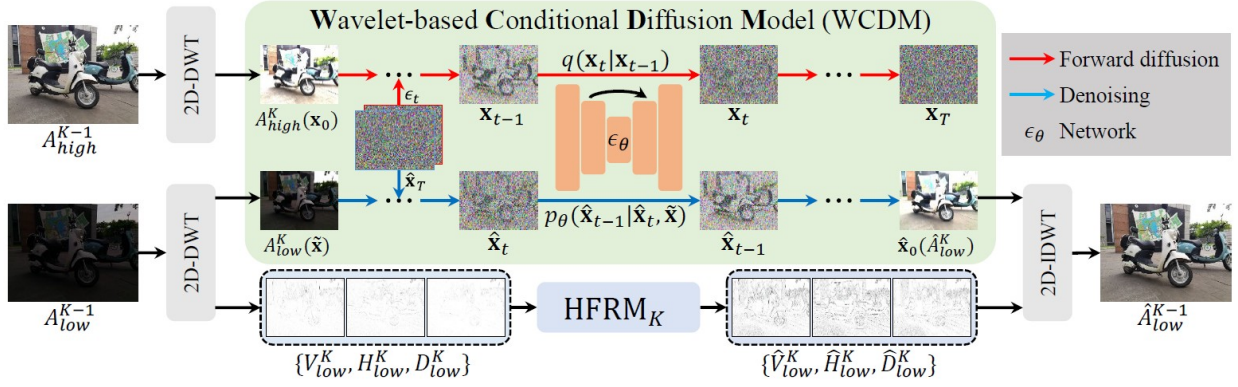


Figure 1: Overall pipeline of baseline. Figure reproduced from DiffLL [2].

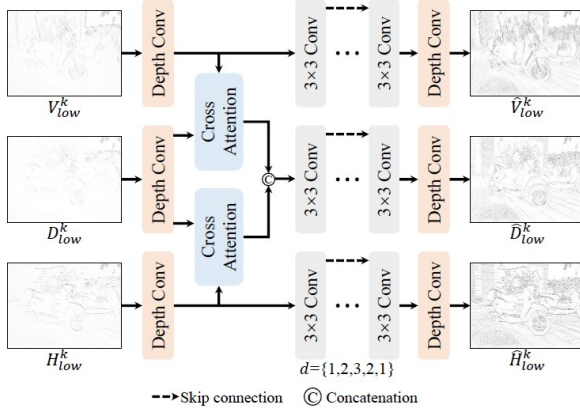


Figure 2: Architecture of the High-Frequency Restoration Module (HFRM). Figure reproduced from DiffLL [2].

components to facilitate the restoration of diagonal details. Subsequently, a series of convolutional layers are applied to enhance and refine the recovered high-frequency information. Finally, the restored low-frequency and high-frequency coefficients are combined through inverse wavelet transformation to reconstruct the enhanced image.

The training objective consists of multiple loss functions. An MSE loss is applied to supervise the predicted diffusion noise, as well as the reconstructed low-frequency and high-frequency coefficients against their corresponding ground-truth counterparts. A Total Variation (TV) loss is further introduced to regularize the restored high-frequency coefficients and suppress undesirable artifacts. In addition, an L1 reconstruction loss and a structural similarity loss, formulated as $1 - \text{SSIM}$, are imposed on the final reconstructed image to improve both pixel-wise fidelity and perceptual quality.

Figure 1 presents the overall pipeline of DiffLL. The low-frequency wavelet coefficients are restored through a diffusion-based network, while the high-frequency coefficients are reconstructed using a

lightweight convolutional network. Figure 2 illustrates the architecture of HFRM.

2.3 Our Method and Novelty

To overcome the performance bottlenecks of the standard DiffLL framework and fully unleash the model’s potential, we extensively explored five technical modifications spanning training curricula, architectural improvements, and inference-phase optimizations. Through rigorous empirical validation, three of these methods—Test-Time Augmentation (TTA), Loss Function Modification, and Two-Level Wavelet Frequency Decomposition—demonstrated substantial efficacy. The remaining two methods—Network Pruning and Wavelet Soft Thresholding—were found to be less effective for this specific task.

The comprehensive breakdown of our methodology and novel contributions is detailed below.

2.3.1 Effective Methods.

1. *Test-Time Augmentation (TTA)*: Standard inference directly processes a single low-light image, which is highly vulnerable to the network’s implicit spatial grid bias. We introduce a robust inference pipeline combining geometric and spatial transformations. First, we implement a Geometric Ensemble (8x TTA) utilizing combinations of 90° rotations and horizontal/vertical flips. Since these operations are mathematically lossless and reversible, they effectively mitigate stochastic high-frequency artifacts. Second, we incorporate Pixel Shifting (sub-pixel spatial translation) as a spatial over-sampling strategy. By shifting the input image slightly prior to inference and reversing the shift on the output, we counteract the translation invariance deficit inherent in convolutional layers, enabling pixel-wise mutual compensation during the final ensemble fusion.

2. *Loss Function Optimization (Modified Loss)*: Standard pixel-wise losses (e.g., L1 or MSE) in low-light restoration often produce over-smoothed results that preserve global structure but lose fine details and natural texture. We adopt a hybrid loss combining three complementary objectives: pixel reconstruction fidelity, structural

similarity, and perceptual realism. The total training loss is formulated as:

$$\mathcal{L}_{\text{total}} = w_{\text{MSE}} \cdot \mathcal{L}_{\text{MSE}} + w_{\text{SSIM}} \cdot \mathcal{L}_{\text{SSIM}} + w_{\text{LPIPS}} \cdot \mathcal{L}_{\text{LPIPS}}$$

where $\mathcal{L}_{\text{MSE}}$ ensures pixel-level reconstruction accuracy via standard L2 distance, $\mathcal{L}_{\text{SSIM}} = 1 - \text{SSIM}(y_{\text{pred}}, y_{\text{gt}})$ captures structural similarity in local windows via luminance, contrast, and structure preservation, and $\mathcal{L}_{\text{LPIPS}}$ measures perceptual distance using deep feature embeddings from a pretrained network (AlexNet/VGG), capturing human-perceived quality better than pixel-space metrics. The weighting strategy is set to $w_{\text{MSE}} = 1.0$, $w_{\text{SSIM}} = 0.1$, and $w_{\text{LPIPS}} = 0.1$, ensuring MSE dominates for optimization stability while SSIM and LPIPS provide secondary guidance toward natural texture and perceptual coherence. This hybrid formulation reduces over-smoothing, improves perceptual quality, and robustly balances metric trade-offs while maintaining training stability. The modified loss was applied uniformly across all model variants (ddpm, ddpm2, ddpm3) to ensure fair comparison in ablation studies.

3. Two-Level Wavelet Frequency Decomposition: While standard DiffLL applies a single-level Haar wavelet transform, it often fails to completely separate highly correlated low-frequency illumination fields from complex high-frequency sensor noise. We propose a Two-Level Wavelet Frequency Decomposition scheme. By recursively applying the Haar wavelet transform to the low-frequency approximation coefficients (LL) for a second level (LL_2), we compress the dimensional footprint of the lighting components. This allows the diffusion core to focus its generative capacity on an even lower-dimensional latent representation, drastically stabilizing the denoising process while reducing computational overhead.

2.3.2 Explored but Ineffective Methods.

1. Data Pruning (Sample Selection via OHM): In training on diverse low-light datasets, not all samples contribute equally to model generalization. Poorly-selected or redundant examples can waste computational budget and introduce noise into gradient updates. We implement Online Hard Example Mining (OHM) as a sample-selection strategy to dynamically prune the training batch during optimization. At each training step, we compute per-sample reconstruction loss (MSE) for the current batch and apply a selection criterion to retain only the most informative subset. Specifically, we explore three selection modes: hardest 50% (samples with highest loss, targeting difficult regions), easiest 50% (samples with lowest loss, emphasizing stable examples), and middle 50% (moderate-difficulty samples, balancing stability and challenge).

2. Wavelet Soft Thresholding: Given that the baseline method utilizes wavelet transforms, we attempted to embed traditional Wavelet Soft Thresholding into the high-frequency coefficients before the HFRM to perform explicit pre-denoising.

3 Experimental results

3.1 Training Setup

3.1.1 Dataset. We train our model on paired low-light image restoration data, where each sample consists of a low-light input image and its corresponding normal-light ground-truth target. The training pipeline follows a supervised setting with separate training,

validation, and test splits. During training, images are cropped into fixed-size patches to improve optimization stability and memory efficiency, while validation and test evaluation are performed on the full benchmark protocol. The validation set contains 1000 images and is used for checkpoint selection and ablation comparison.

3.1.2 Implementation Details. Our framework is implemented in PyTorch. Unless otherwise specified, the model is trained on 256×256 image patches using the proposed wavelet-guided diffusion restoration pipeline. We evaluate the effect of several incremental modifications, including loss function optimization, architectural refinement, and enhanced test-time augmentation (TTA). In particular, the modified architecture incorporates stronger wavelet-frequency modeling, while the modified loss combines pixel fidelity, structural similarity, and perceptual supervision. For model efficiency, the final submitted model has a parameter size of 22.079M and computational cost of 43.9145 GFLOPs.

3.1.3 Evaluation Metrics. We evaluate restoration performance using three image-quality metrics: PSNR, SSIM, and LPIPS. PSNR measures pixel-level reconstruction fidelity, SSIM evaluates structural consistency, and LPIPS measures perceptual similarity in deep feature space. Since practical deployment is also important, we additionally report model size (in millions of parameters) and FLOPs (in giga floating-point operations). Therefore, a better model should achieve higher PSNR and SSIM, lower LPIPS, and lower complexity whenever possible.

3.2 Experimental Results on the Test Set

On the test set, our final model achieves **21.3335** PSNR, **0.6515** SSIM, and **0.4393** LPIPS, with a model size of **22.079M** parameters and **43.9145** GFLOPs. These results indicate that the proposed method achieves competitive restoration quality while maintaining a practical computational budget.

3.3 Experimental Results on the ValidationSet

Baseline at 220 epochs: The baseline model trained for 220 epochs achieves PSNR = 21.1682, SSIM = 0.6327, and LPIPS = 0.4626. This serves as the reference point for subsequent modifications.

After loss modification and architecture refinement (47 epochs): After replacing the original loss with the hybrid loss and introducing the architectural refinement, the validation performance becomes PSNR = 21.1845, SSIM = 0.6283, and LPIPS = 0.3578. Compared with the baseline, this setting significantly improves LPIPS, showing that the modified loss and architecture effectively enhance perceptual quality and suppress over-smoothed outputs, although SSIM decreases slightly.

After adding rotation-based TTA: When rotation-based TTA is incorporated, the validation performance further changes to PSNR = 21.2003, SSIM = 0.6298, and LPIPS = 0.3653. This result shows that geometric ensemble inference provides a small improvement in reconstruction fidelity and structural consistency, while maintaining strong perceptual quality.

After adding pixel shifting TTA: Finally, after introducing pixel shifting into the TTA pipeline, the validation performance improves to PSNR = 21.3524, SSIM = 0.6376, and LPIPS = 0.3639. This is the

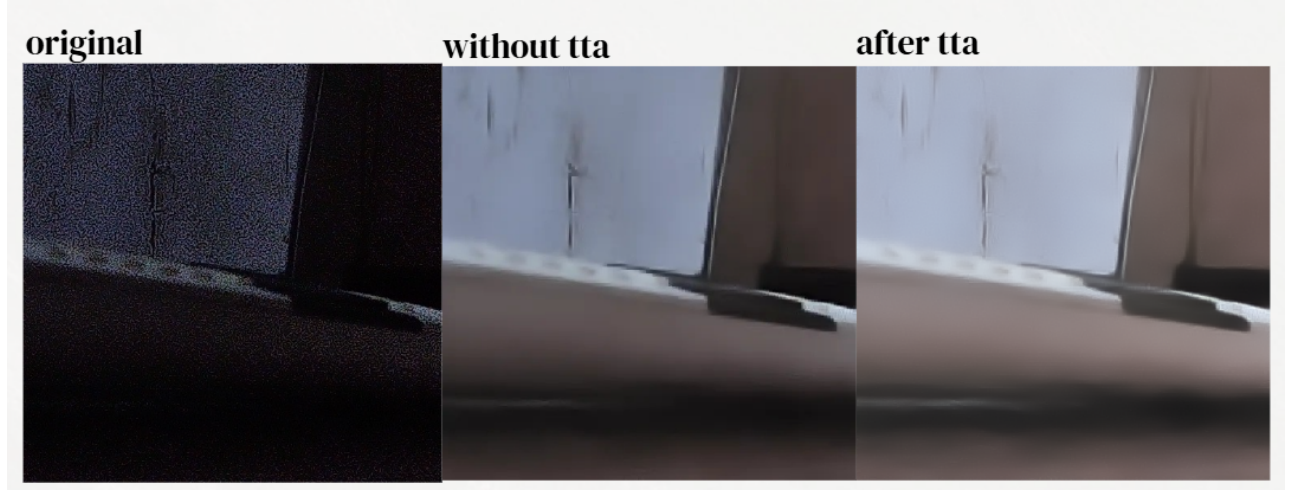


Figure 3: The result of our model

best overall trade-off among the ablation settings, indicating that pixel shifting helps reduce spatial bias and enhances ensemble robustness during inference.

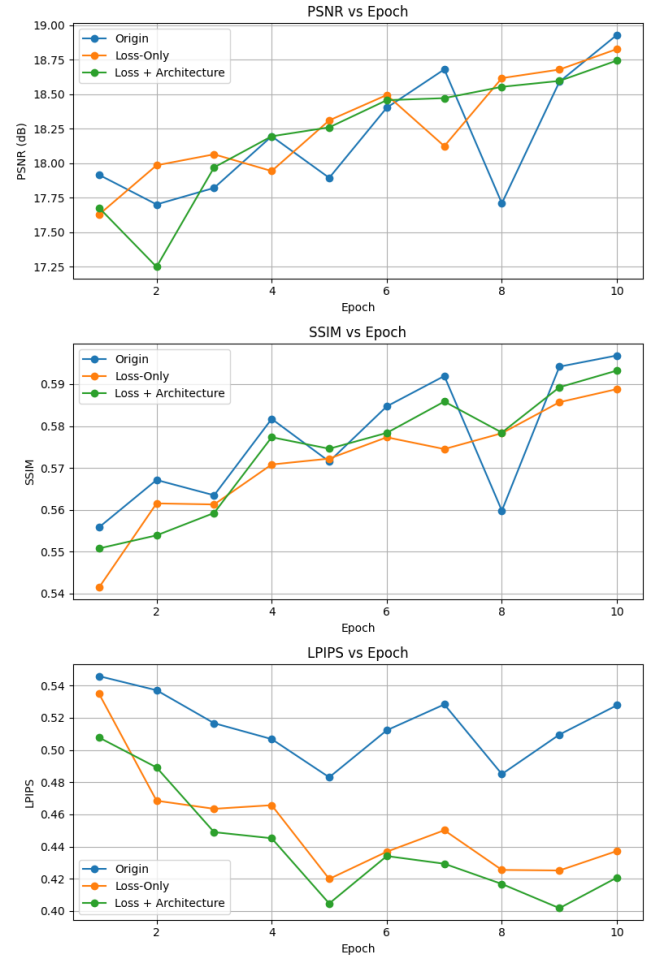


Figure 4: Comparison of loss function optimization and Two-Level Wavelet Frequency Decomposition

3.4 Ablation Studies

3.4.1 Ablation Analysis of Model Components. To further dissect the individual contributions of the modified loss function and the two-level wavelet frequency decomposition, we track the convergence trajectories of three distinct configurations across the initial 10 training epochs, as illustrated in Fig. 4. The models compared are the baseline framework (Blue Line), the framework with our modified loss function (Orange Line), and our complete configuration combining both the two-level wavelet decomposition and the modified loss (Green Line, Ours).

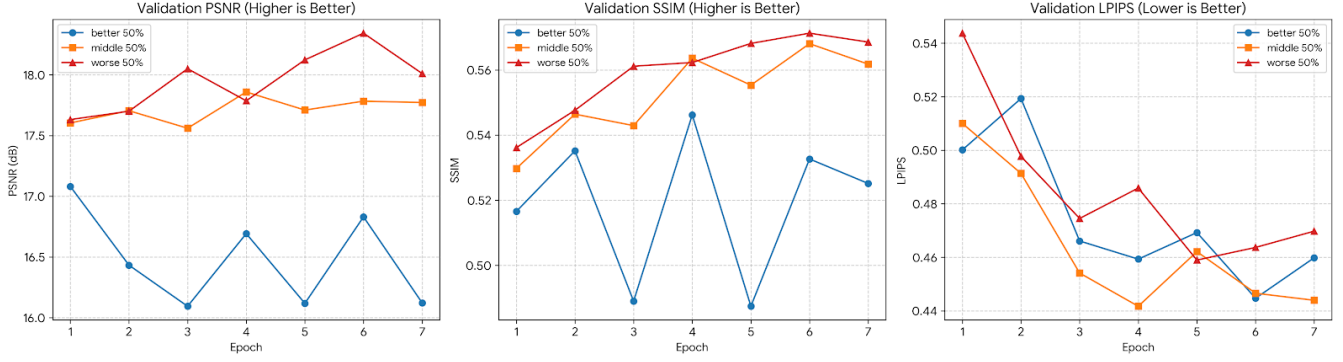


Figure 5: Data pruning from epoch 1 to 7

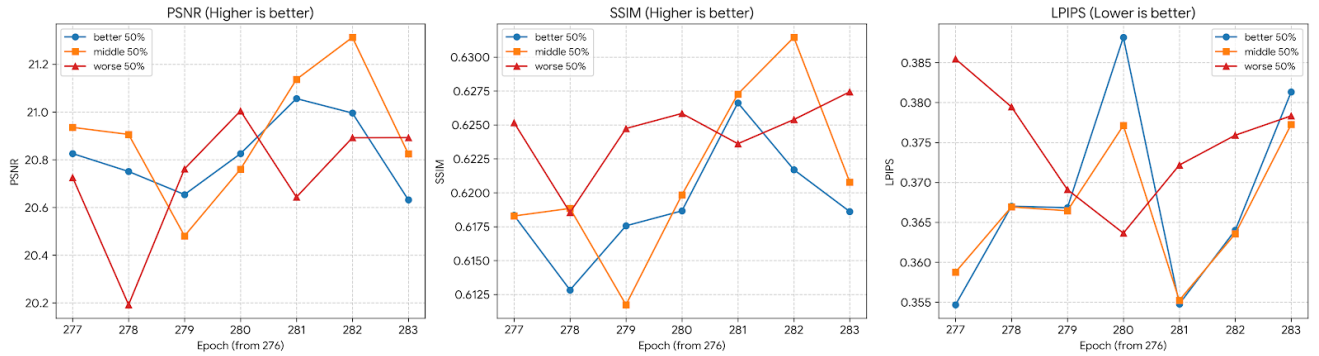


Figure 6: Data pruning from epoch 277 to 283

Effect of Two-Level Wavelet Frequency Decomposition: A direct comparison between the Orange Line (Modified Loss) and the Green Line (Ours) from Epoch 0 to 10 demonstrates that the introduction of the Two-Level Wavelet Frequency Decomposition substantially improves the stability of the PSNR trajectory. Furthermore, this recursive multi-level decomposition consistently yields superior performance across both SSIM and LPIPS metrics. By further decomposing the low-frequency illumination approximation, the core network is insulated from highly correlated background fields, leading to highly stabilized structural reconstruction and better convergence behaviors.

Effect of Loss Function Optimization: Comparing the Blue Line (Baseline) with the Orange Line (Modified Loss) reveals that our optimization of the training objective offers distinct advantages in capturing perceptual fidelity. Specifically, the integration of the structural similarity regularizer ($1 - \text{SSIM}$) alongside the high-frequency Total Variation (TV) regularizer significantly accelerates the reduction of the LPIPS metric. This indicates that while standard pixel-wise constraints tend to over-smooth localized patterns, our compound loss actively forces the network to preserve sharp structural boundaries and suppress generative artifacts, aligning the final outputs more closely with human visual perception.

3.4.2 Effect of Data Pruning (Curriculum Filtering): As illustrated in the trajectory from Epoch 0 to 7 in Fig. 5, utilizing the *Better*

50% data partition (retaining only low-loss, easy samples) exhibited the worst convergence and lowest peak performance. Conversely, when observing the fine-tuning trajectory from the pre-converged state (Epoch 277 to 283) in Fig. 6, the *Worse 50%* partition (focusing entirely on hard, high-loss samples) demonstrated a distinctly superior recovery capability. Notably, the *Middle 50%* configuration maintained relatively stable and balanced results across both training windows.

Although all three hard-threshold pruning strategies caused a baseline performance degradation compared to the unpruned distribution, these findings offer critical insights for hyperparameter tuning. Instead of a rigid 50% split, future configurations could dynamically expand the mid-range distribution density to 90% (*Middle 90%*). Theoretically, this adaptive threshold tuning holds the potential to precisely filter out highly corrupted outliers (the extreme worst) and saturated patches that no longer yield informative gradients (the extreme best), thereby creating an optimal sample manifold for stable generative fine-tuning.

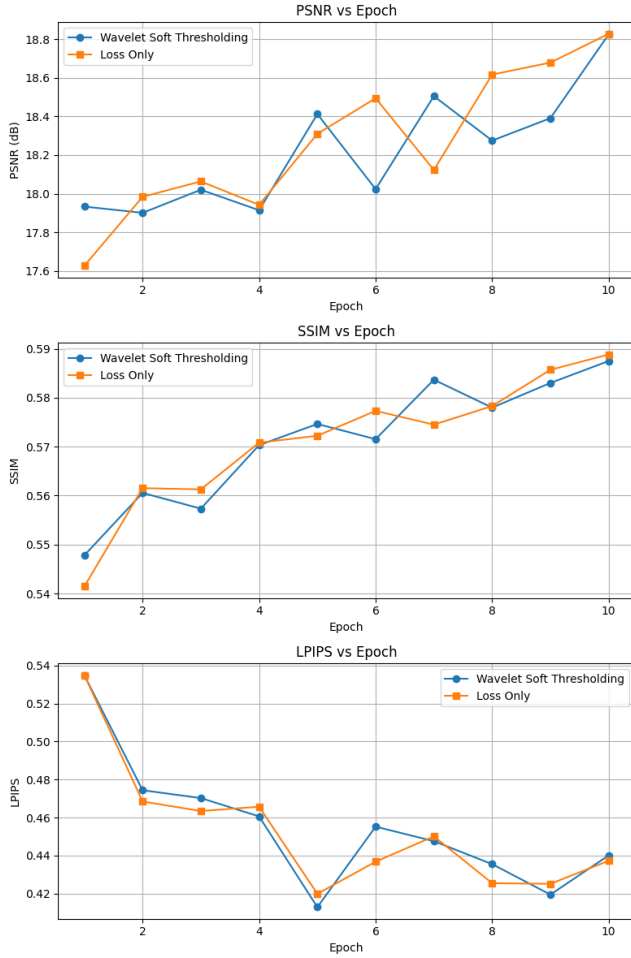


Figure 7: Comparison of Wavelet Soft Thresholding

3.4.3 Effect of Wavelet Soft Thresholding: As observed from the convergence curves in Fig. 7, the integration of traditional Wavelet Soft Thresholding failed to yield measurable quantitative improvements and was largely ineffective for this joint restoration task. Empirically, embedding fixed thresholding operators within the high-frequency wavelet coefficients introduced an adverse smoothing effect. We attribute this degradation to the indiscriminate nature of static thresholds; while they suppress baseline sensor noise, they inadvertently discard low-amplitude, non-redundant structural signals. Consequently, this rigid filtering blurred genuine high-frequency details and low-contrast edges, which fundamentally starved the downstream attention-driven High-Frequency Restoration Module (HFRM) of critical geometric cues needed for adaptive texture reconstruction.

4 Conclusion

In this project, we successfully extended and optimized the DiffLL framework for joint image denoising and low-light enhancement. Through extensive ablation studies spanning both the training

curriculum and the test-time inference pipeline, we reached several critical conclusions regarding generative restoration models:

First, we demonstrated that **Test-Time Augmentation (TTA) integrated with Pixel Shifting** serves as an exceptionally powerful mechanism to suppress stochastic artifacts and generative noise inherent in diffusion model outputs. By aggregating multi-view and sub-pixel shifted predictions, the ensemble effectively smooths out random high-frequency fluctuations while preserving structural fidelity, establishing a peak benchmark of 21.3524 dB in PSNR.

Second, our frequency-domain investigations revealed that **selectively preserving moderate low-frequency details** yields far superior image reconstruction compared to raw or overly compressed representations. This optimal low-frequency guidance stabilizes the reverse diffusion process, ensuring uniform luminance enhancement without inducing over-exposure or structural distortion.

Third, regarding efficiency optimizations, our preliminary experiments with **Data and Network Pruning** led to minor performance drop-offs, indicating that the network relies heavily on dense representations to invert severe degradations. However, we conclude that pruning cannot be definitively ruled out; it requires a more meticulous hyperparameter grid search and adaptive threshold tuning to properly balance the trade-off between computational compression and restoration quality.

Finally, we empirically verified that traditional **Wavelet Soft Thresholding is unsuitable** for this deep generative task. The hard-coded, static thresholding mechanism tends to aggressively strip away genuine edge textures alongside actual noise, which directly conflicts with and disrupts the highly adaptive, attention-driven high-frequency restoration module (HFRM) within the DiffLL architecture.

In summary, our final optimized configuration underscores the necessity of coupling deep generative priors with robust, transformation-invariant inference strategies to push the boundaries of low-light image restoration.

Table 1: Team Member Contributions and Responsibilities

Student ID	Contribution
112062139 Wei-Cheng Liu	Research on the baseline model, conceptualization of novel methods, and report presentation.
112062117 Jheng-Jie Wang	Model training execution, methodology conceptualization, presentation preparation, and quantitative data analysis.
112062209 Geng-An Hu	Model training execution, presentation slide (PPT) design, and conceptualization of novel methods.
112062320 Shao-Pu Hsieh	Research on the baseline model, presentation slide (PPT) design, and conceptualization of novel methods.

References

- [1] Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising Diffusion Probabilistic Models. In Advances in Neural Information Processing Systems (NeurIPS), Vol. 33. 6840–6851.
- [2] Hai Jiang, Ao Luo, Haoqiang Fan, Songchen Han, and Shuaicheng Liu. 2023. Low-light image enhancement with wavelet-based diffusion models. ACM Transactions on Graphics (TOG) 42, 6 (2023), 1–14.
- [3] Yifan Jiang, Xinyu Gong, Ding Liu, Yu Cheng, Chen Fang, Xiaohui Shen, Jianchao Yang, Pan Zhou, and Zhangyang Wang. 2021. EnlightenGAN: Deep Light Enhancement Without Paired Supervision. IEEE Transactions on Image Processing 30 (2021), 2340–2349.
- [4] Jialun Peng, Dong Liu, Songcen Xu, and Houqiang Li. 2021. Generating Diverse Structure for Image Inpainting with Hierarchical VQ-VAE. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). 10775–10784.
- [5] Divya Shanmugam, Davis W. Blalock, Guha Balakrishnan, and John V. Guttag. 2020. When and Why Test-Time Augmentation Works. CoRR abs/2011.11156 (2020). arXiv:2011.11156 <https://arxiv.org/abs/2011.11156>
- [6] Chen Wei, Wenjing smokes Wang, Wenhan Yang, and Jiaying Liu. 2018. Deep Retinex Decomposition for Low-Light Enhancement. In Proceedings of the British Machine Vision Conference (BMVC).

References

- [1] Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising Diffusion Probabilistic Models. In Advances in Neural Information Processing Systems (NeurIPS), Vol. 33. 6840–6851.
- [2] Hai Jiang, Ao Luo, Haoqiang Fan, Songchen Han, and Shuaicheng Liu. 2023. Low-light image enhancement with wavelet-based diffusion models. ACM Transactions on Graphics (TOG) 42, 6 (2023), 1–14.
- [3] Yifan Jiang, Xinyu Gong, Ding Liu, Yu Cheng, Chen Fang, Xiaohui Shen, Jianchao Yang, Pan Zhou, and Zhangyang Wang. 2021. EnlightenGAN: Deep Light Enhancement Without Paired Supervision. IEEE Transactions on Image Processing 30 (2021), 2340–2349.
- [4] Jialun Peng, Dong Liu, Songcen Xu, and Houqiang Li. 2021. Generating Diverse Structure for Image Inpainting with Hierarchical VQ-VAE. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). 10775–10784.
- [5] Divya Shanmugam, Davis W. Blalock, Guha Balakrishnan, and John V. Guttag. 2020. When and Why Test-Time Augmentation Works. CoRR abs/2011.11156 (2020). arXiv:2011.11156 <https://arxiv.org/abs/2011.11156>
- [6] Chen Wei, Wenjing smokes Wang, Wenhan Yang, and Jiaying Liu. 2018. Deep Retinex Decomposition for Low-Light Enhancement. In Proceedings of the British Machine Vision Conference (BMVC).